

A Production Economy

RBC structure, rational expectations and Dynare

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SESSION OBJECTIVE

- ▶ Build the core equilibrium conditions of a stochastic production economy.
- ▶ See how rational-expectations models are solved and diagnosed.
- ▶ Translate an economic model into readable Dynare code.

Handout: Session 2 **Code:** `code/session2/`

PLAN

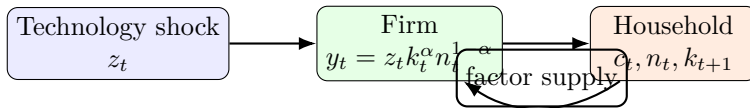
① Production economy

② Solution

③ Dynare

④ Takeaways

THE RBC BENCHMARK



Resource constraint: $c_t + i_t = y_t$, with $k_{t+1} = (1 - \delta)k_t + i_t$.

EQUILIBRIUM CONDITIONS ORGANIZE THE MODEL

Household

$$u_c(c_t, n_t) = \beta \mathbb{E}_t \{ u_c(c_{t+1}, n_{t+1}) (r_{t+1} + 1 - \delta) \}$$

$$-u_n(c_t, n_t) = u_c(c_t, n_t) w_t$$

Firm

$$r_t = \alpha z_t k_t^{\alpha-1} n_t^{1-\alpha}$$

$$w_t = (1 - \alpha) z_t k_t^{\alpha} n_t^{-\alpha}$$

Discipline: every variable needs an equation, a timing convention and a unit.

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- 2 **Solution**
- 3 Dynare
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LINEAR RATIONAL EXPECTATIONS FORM

After approximation, many models take the form

$$A \mathbb{E}_t x_{t+1} = Bx_t + Cx_{t-1} + D\varepsilon_t.$$

- ▶ Predetermined states carry history.
- ▶ Forward-looking controls respond to expected future states.
- ▶ A unique stable solution requires the appropriate number of unstable roots.

WHAT AN IMPULSE RESPONSE CAN – AND CANNOT – TELL US

- ▶ An IRF is a conditional model experiment, not a causal estimate by itself.
- ▶ Its shape reflects parameter values, shock normalization and solution method.
- ▶ Compare mechanisms, not only peak responses.

Checklist

Sign? Units? Horizon? Shock size? Stability? Plausible transmission?

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A READABLE DYNARE SKELETON

```
var c k y n z;  
varexo e;  
parameters beta alpha delta rho sigma_e;  
  
model;  
  y = exp(z)*k(-1)^alpha*n^(1-alpha);  
  c + k = y + (1-delta)*k(-1);  
  1/c = beta*(1/c(+1))*(alpha*y(+1)/k);  
  z = rho*z(-1) + e;  
end;
```

Rule: code mirrors the economic model; comments document transformations and timing.

DYNARE WORKFLOW



- ▶ Read Dynare messages; do not treat warnings as decorative.
- ▶ Generated folders and output are reproducible and intentionally ignored by Git.

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TAKEAWAYS AND PREPARATION

- ▶ The RBC model connects preferences, technology and resource feasibility.
- ▶ Rational expectations impose solution and determinacy requirements.
- ▶ Dynare is valuable when the code remains a transparent representation of the model.

Before Session 3: run `basicRBC.mod`; retain the code, not generated outputs.